

DSA STUDY BLUEPRINT SERIES

17. Sorting Algorithms Bubble Selection Insertion

Bubble, Selection, and Insertion Sort Implementations

Section 03 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Selection Sort:** Repeatedly places minimum element at start.
- **Bubble Sort:** Swaps adjacent out-of-order pairs.
- **Insertion Sort:** Inserts elements into sorted prefix in-place.

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
// Selection Sort
for (int i = 0; i < n-1; i++) {
    int minIdx = i;
    for (int j = i+1; j < n; j++)
        if (arr[j] < arr[minIdx]) minIdx = j;
    swap(arr[i], arr[minIdx]);
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Selection Sort	$O(N^2)$ time, $O(1)$ space
Bubble/Insertion	$O(N^2)$ time / $O(N)$ best case

Key Practices & Gotchas

- **Important:** Insertion sort is highly efficient for small or partially sorted datasets.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's dry run a pass of Selection Sort on array `[35, 22, 12, 45]` :

Pass Index	Unsorted Prefix Range	Scan result: Minimum Element	Swap Action	Array State after Pass
i = 0	[35, 22, 12, 45]	Min is 12 (at index 2)	Swap index 0 and 2	<code>[12, 22, 35, 45]</code>

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Stability:** Insertion Sort is stable (preserves relative duplicate order), whereas Selection Sort is unstable due to long-range element swaps.
- **Adaptive Property:** Insertion Sort runs in $O(N)$ best case for nearly-sorted arrays.
- **Related LeetCode Problems:** #912 Sort an Array, #75 Sort Colors, #273 Regular Array Sorting checks.